MATH 111 Calculus 1, NUU, FALL 2026

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Continuity (Section 1.8) and limits at infinity (Section 3.4): the definition of continuity, the three ways it fails, building continuous functions, the Intermediate Value Theorem, horizontal asymptotes and the behaviour of a function far out.

Key facts to have at hand

- **Continuity at a number a:** $\lim_{x\to a} f(x) = f(a)$. Three things are hidden in this one equation: $f(a)$ is defined; the limit exists; the two are *equal*. If f is defined near a but not continuous at a, it is **discontinuous** at a.
- **Three ways to fail:** a *removable* discontinuity (the limit exists but the value is missing or wrong — one value repairs it); a *jump* (the one-sided limits exist and differ — no value repairs it); an *infinite* discontinuity (a one-sided limit is $\pm\infty$ — no value repairs it).
- **One-sided continuity:** continuous from the right at a means $\lim_{x\to a^+} f(x) = f(a)$; from the left, $\lim_{x\to a^-} f(x) = f(a)$. **Continuous on an interval** means continuous at every number inside it, and one-sidedly at any endpoint that belongs to it.
- **Building continuous functions:** if f, g are continuous at a then so are $f \pm g$, cf, fg and f/g *provided* $g(a) \neq 0$; if g is continuous at a and f at $g(a)$, then $f \circ g$ is continuous at a. A limit passes inside a continuous function: $\lim_{x\to a} f(g(x)) = f\big(\lim_{x\to a} g(x)\big)$.
- **Which functions:** polynomials, rational, root and trigonometric functions are continuous **on their domains** — and there "limit" means "substitute".
- **Intermediate Value Theorem:** if f is continuous on the *closed* interval $[a, b]$ and N is any number between $f(a)$ and $f(b)$ with $f(a) \neq f(b)$, then there is a number c in (a, b) with $f(c) = N$. It promises *at least one* such c and says nothing about where it is. Without continuity it is false.
- **Limits at infinity:** $\lim_{x\to\infty} f(x) = L$ means $f(x)$ is as close to L as we like for all x large enough. Then $y = L$ is a **horizontal asymptote** — which the graph is allowed to cross, even infinitely often.
- **The basic limit:** $\lim_{x\to\pm\infty} \frac{1}{x} = 0$, and $\lim_{x\to\infty} \frac{1}{x^r} = 0$ for every $r > 0$. All the limit laws hold for $x \to \pm\infty$ as they do for $x \to a$.
- **Method for a quotient:** divide the numerator and the denominator by the *highest power of x in the denominator*, then use $1/x^r \to 0$. For roots remember $\sqrt{x^2} = |x|$, so $\sqrt{x^2} = -x$ when $x < 0$: the sign is your job, not the algebra's.
- **$\infty - \infty$ and ∞/∞ are instructions to rewrite**, like $0/0$: divide, or multiply by the conjugate. $\lim_{x\to\infty} f(x) = \infty$ means $f(x)$ exceeds every number for x large enough.
- **Precise definition at infinity:** $\lim_{x\to\infty} f(x) = L$ means: for every $\varepsilon > 0$ there is a number N such that if $x > N$ then $|f(x) - L| < \varepsilon$. Same grammar as ε–δ: ε is the challenge, N the response.

Part I · Continuity and the Intermediate Value Theorem (Section 1.8) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is continuous at a, then $\lim_{x\to a} f(x) = f(a)$.
2. If $\lim_{x\to a} f(x)$ exists, then f is continuous at a.
3. If $f(a)$ is defined and $\lim_{x\to a} f(x)$ exists, then f is continuous at a.
4. $f(x) = 1/x$ is a continuous function.
5. $f(x) = 1/x$ is continuous at 0.
6. If f and g are continuous at a, then f/g is continuous at a.
7. If $f(1) = -2$ and $f(3) = 5$, then $f(c) = 0$ for some c in $(1, 3)$.

8. If f is continuous on $[1,3]$, $f(1) = -2$ and $f(3) = 5$, then f has *exactly one* zero in $(1,3)$.
9. If f has a jump discontinuity at a, one can change the value $f(a)$ to make f continuous at a.
10. $f(x) = |x|$ is continuous at 0.

Solution.

1 True. This is the definition of continuity at a number, word for word.

2 False. A limit can exist where the function has no value at all. For $f(x) = \dfrac{x^2-4}{x-2}$ we have $\lim_{x\to 2} f(x) = 4$, but $f(2)$ is undefined, so the first of the three conditions fails.

3 False. Both may be present and still disagree. Let $f(x) = \dfrac{x^2-4}{x-2}$ for $x \neq 2$ and $f(2) = 7$: the limit is 4, the value is 7, and continuity needs them equal.

4 True. "f is a continuous function" means: continuous at every number *of its domain*. The domain of $1/x$ is $\{x : x \neq 0\}$, and at every such number the quotient law applies. There is nothing to check at 0, because 0 is not in the domain.

5 False. Continuity *at* 0 would require $f(0)$ to exist first, and it does not. (What happens there is an infinite discontinuity: $1/x \to +\infty$ from the right and $-\infty$ from the left.)

6 False in general. The quotient law needs $g(a) \neq 0$. Take $f(x) = 1$ and $g(x) = x$ at $a = 0$: both are continuous at 0, but $f/g = 1/x$ is not even defined there.

7 False. Nothing was said about continuity, and the Intermediate Value Theorem is false without it. The step function $f(x) = -2$ for $x < 2$ and $f(x) = 5$ for $x \geq 2$ satisfies $f(1) = -2$, $f(3) = 5$ and never takes the value 0.

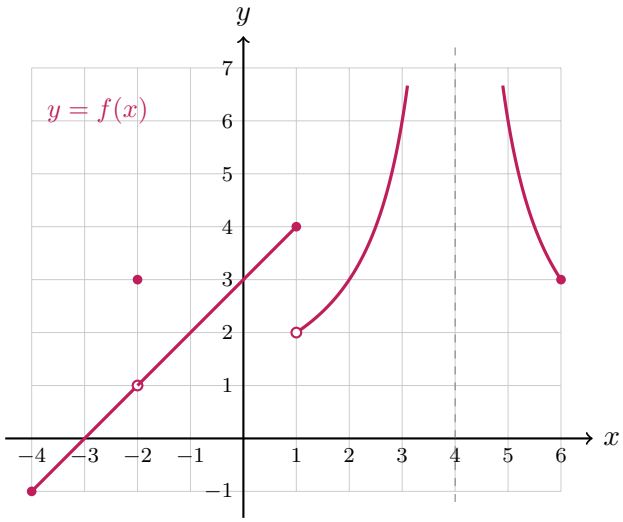
8 False. The IVT promises *at least one* c; it never says how many. A continuous function can cross the level 0 three times, or infinitely often, between 1 and 3.

9 False. At a jump the two one-sided limits exist and are *different*. Continuity at a needs $f(a)$ to equal both of them at once, which no number can do. Only a *removable* discontinuity can be repaired by a value.

10 True. For $x > 0$, $|x| = x \to 0$; for $x < 0$, $|x| = -x \to 0$. Both one-sided limits are 0, so $\lim_{x\to 0} |x| = 0$, and $|0| = 0$: limit and value agree.

Problem 1 *Reading continuity off a graph.*

The graph of a function f is shown; the dashed line is a vertical asymptote. (f is defined at every number of $[-4, 6]$ except $x = 4$.)



Find each of the following, or say that it does not exist.

(a) $\lim_{x\to -2^-} f(x)$ and $\lim_{x\to -2} f(x)$

(b) $f(-2)$

(c) $\lim_{x\to 1^-} f(x)$ and $\lim_{x\to 1^+} f(x)$

(d) $\lim_{x\to 4} f(x)$

(e) At which numbers is f discontinuous? Name the type of each discontinuity.

(f) At which of them can a change of one value make f continuous, and what value? On which intervals is f continuous?

Solution.

(a) Both are 1. Coming from the left along the segment, the values climb towards the open dot at $(-2, 1)$; coming from the right they descend to the same open dot. The two one-sided limits agree, so the two-sided limit exists and equals 1. The limit never looks at the point itself, so the filled dot higher up is irrelevant here.

(b) $f(-2) = 3$: the value is where the *filled* dot is.

(c) From the left the segment rises to the filled dot at $(1, 4)$, so $\lim_{x\to 1^-} f(x) = 4$. From the right the next branch starts at the open dot at $(1, 2)$, so $\lim_{x\to 1^+} f(x) = 2$. They differ, so $\lim_{x\to 1} f(x)$ does not exist.

(d) As $x \to 4$ from either side the graph runs up the dashed line without bound: $\lim_{x\to 4} f(x) = +\infty$. This is a statement about behaviour, not a number — the limit does not exist as a real number, and $x = 4$ is a vertical asymptote.

(e) Three points, one of each kind. At $x = -2$ the limit exists (1) but the value is different (3): a *removable* discontinuity. At $x = 1$ the one-sided limits exist and differ (4 and 2): a *jump*. At $x = 4$ a one-sided limit is infinite: an *infinite* discontinuity. Everywhere else the graph is drawn in one unbroken stroke.

(f) Only the removable one can be repaired, and only in one way: define $f(-2) = 1$, the value of the limit — then limit and value agree. A jump cannot be repaired, because no single number equals both 4 and 2; an infinite discontinuity cannot be repaired either, because there is no limit to match. On the intervals $[-4, -2)$, $(-2, 1]$, $(1, 4)$ and $(4, 6]$ the function is continuous (at -4 and at 6 one-sidedly, which is all an endpoint can be asked for).

Problem 2 *Where is each function continuous? Give the largest set, and name the theorem you use.*

(a) $f(x) = \dfrac{x^2 - 9}{x^2 - 4x + 3}$

(b) $g(x) = \sqrt{5 - x} + \sqrt{x + 1}$

(c) $h(x) = \dfrac{x + 2}{\sqrt{x^2 - 4}}$

(d) $k(x) = \sin\!\left(\dfrac{1}{x - 2}\right)$

(e) $m(x) = \dfrac{\cos x}{x^2 + 1}$

Solution.

(a) f is a *rational function*, and a rational function is continuous on its domain. The domain excludes the zeros of the denominator: $x^2 - 4x + 3 = (x - 1)(x - 3) = 0$ at $x = 1$ and $x = 3$. So f is continuous on $\mathbb{R} \setminus \{1, 3\} = (-\infty, 1) \cup (1, 3) \cup (3, \infty)$. (It is worth noticing that the two excluded points are not alike: at 3 the numerator also vanishes, $\frac{(x-3)(x+3)}{(x-1)(x-3)} = \frac{x+3}{x-1} \to 3$, a hole; at 1 the quotient blows up. Neither is in the domain, so neither belongs in the answer.)

(b) Both $\sqrt{5 - x}$ and $\sqrt{x + 1}$ are root functions composed with polynomials, hence continuous wherever they are defined; a sum of continuous functions is continuous. We need $5 - x \geq 0$, i.e. $x \leq 5$, and $x + 1 \geq 0$, i.e. $x \geq -1$. So g is continuous on $[-1, 5]$ — at the endpoints one-sidedly, which is what "continuous on a closed interval" means.

(c) The numerator is a polynomial and the denominator a root function, so the quotient is continuous wherever the denominator is defined *and non-zero*: we need $x^2 - 4 > 0$ (not merely ≥ 0), that is $|x| > 2$. So h is continuous on $(-\infty, -2) \cup (2, \infty)$.

(d) Write $k = f \circ g$ with $g(x) = \dfrac{1}{x - 2}$ and $f(u) = \sin u$. Here g is continuous at every $x \neq 2$ and f is continuous everywhere, so by the composition theorem k is continuous at every $x \neq 2$: on $\mathbb{R} \setminus \{2\}$. (At 2 nothing can be repaired: the values oscillate between -1 and 1 for ever, so there is no limit.)

(e) $\cos x$ is continuous on $\mathbb{R}$ and $x^2 + 1$ is a polynomial, continuous on $\mathbb{R}$ and never zero (it is at least 1). By the quotient theorem m is continuous on all of $\mathbb{R}$.

Problem 3 *Classify the discontinuity at the given number as removable, jump or infinite. If it is removable, give the value that repairs it.*

(a) $f(x) = \dfrac{x^2 - 16}{x - 4}$ at $a = 4$

(b) $f(x) = \dfrac{|x - 5|}{x - 5}$ at $a = 5$

(c) $f(x) = \dfrac{x + 2}{x^2 - 4}$ at $a = 2$

(d) $f(x) = \dfrac{x + 2}{x^2 - 4}$ at $a = -2$

(e) $f(x) = \dfrac{\sqrt{x} - 3}{x - 9}$ at $a = 9$

Solution.

(a) Removable. Factor: $x^2 - 16 = (x - 4)(x + 4)$. Dividing by $x - 4$ is allowed only where $x - 4 \neq 0$, that is for $x \neq 4$ — and the limit at 4 uses only such x. So $\lim\limits_{x \to 4} \dfrac{x^2 - 16}{x - 4} = \lim\limits_{x \to 4} \dfrac{(x - 4)(x + 4)}{x - 4} = \lim\limits_{x \to 4} (x + 4) = 8$. The limit exists and the value is missing, which is exactly a removable discontinuity; setting $f(4) = 8$ repairs it.

(b) Jump. For $x > 5$ we have $|x - 5| = x - 5$, so $f(x) = 1$; for $x < 5$ we have $|x - 5| = -(x - 5)$, so $f(x) = -1$. Hence $\lim_{x \to 5^+} f(x) = 1$ and $\lim_{x \to 5^-} f(x) = -1$. Both one-sided limits exist and differ: a jump, and no choice of $f(5)$ can equal both.

(c) Infinite. For $x \neq \pm 2$, $f(x) = \dfrac{x + 2}{(x - 2)(x + 2)} = \dfrac{1}{x - 2}$. As $x \to 2^+$ the denominator is small and positive, so $f(x) \to +\infty$; as $x \to 2^-$ it is small and negative, so $f(x) \to -\infty$. A one-sided limit is infinite, so the discontinuity is infinite and $x = 2$ is a vertical asymptote.

(d) Removable. The same simplification gives $f(x) = \dfrac{1}{x-2}$ for $x \neq \pm 2$, and now $\lim\limits_{x\to -2} \dfrac{1}{x-2} = \dfrac{1}{-4} = -\dfrac{1}{4}$, a perfectly ordinary number. The function has no value at -2 (the original formula gives $0/0$), so the discontinuity is removable: set $f(-2) = -\frac{1}{4}$. Notice that the same formula produces two different kinds of failure at its two bad points — which is why "denominator zero" is where to look, not the verdict.

(e) Removable. Multiply by the conjugate $\sqrt{x}+3$, allowed for $x \geq 0$ since $\sqrt{x}+3 > 0$: $\dfrac{\sqrt{x}-3}{x-9} = \dfrac{(\sqrt{x}-3)(\sqrt{x}+3)}{(x-9)(\sqrt{x}+3)} = \dfrac{x-9}{(x-9)(\sqrt{x}+3)} = \dfrac{1}{\sqrt{x}+3}$ $(x \neq 9)$. Hence the limit is $\dfrac{1}{\sqrt{9}+3} = \dfrac{1}{6}$, and setting $f(9) = \dfrac{1}{6}$ repairs the gap.

Problem 4 *Piecewise functions: find the constant that makes f continuous everywhere.*

(a) $f(x) = \begin{cases} 3x + c, & x \leq 1, \\ x^2 - 2, & x > 1 \end{cases}$

(b) $f(x) = \begin{cases} cx^2, & x < 2, \\ x + 4, & x \geq 2 \end{cases}$

(c) $f(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x \neq 1, \\ c, & x = 1 \end{cases}$

(d) $f(x) = \begin{cases} x + a, & x < 0, \\ x^2 + b, & 0 \leq x \leq 2, \\ 5x - a, & x > 2 \end{cases}$ (find a and b)

(e) $f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0, \\ c, & x = 0 \end{cases}$

Solution.

(a) Each branch is a polynomial, so f is automatically continuous at every $x \neq 1$; only the seam $x = 1$ can go wrong. There, $\lim_{x\to 1^-} f(x) = 3 \cdot 1 + c = 3 + c$ (the branch used for $x \leq 1$) and $\lim_{x\to 1^+} f(x) = 1^2 - 2 = -1$ (the branch used for $x > 1$), while $f(1) = 3 + c$. Continuity needs all three equal: $3 + c = -1$, so $\boxed{c = -4}$.

(b) Only $x = 2$ needs checking. $\lim_{x\to 2^-} f(x) = c \cdot 2^2 = 4c$, and $\lim_{x\to 2^+} f(x) = 2 + 4 = 6 = f(2)$. Setting $4c = 6$ gives $\boxed{c = \frac{3}{2}}$.

(c) For $x \neq 1$, dividing by $x - 1$ is allowed (that is exactly where $x - 1 \neq 0$), so $\dfrac{x^2-1}{x-1} = \dfrac{(x-1)(x+1)}{x-1} = x + 1$ and the limit at 1 is 2. The value at 1 is c, so continuity forces $\boxed{c = 2}$: the removable discontinuity is repaired by the value of the limit, as always.

(d) Two seams, so two equations. At $x = 0$: from the left $x + a \to a$; from the right $x^2 + b \to b$; the value is $f(0) = 0 + b = b$. So $a = b$. At $x = 2$: from the left $x^2 + b \to 4 + b$, which is also $f(2)$; from the right $5x - a \to 10 - a$. So $4 + b = 10 - a$, that is $a + b = 6$. Together with $a = b$ this gives $2a = 6$, so $\boxed{a = b = 3}$.

(e) Away from 0 the quotient of continuous functions is continuous (the denominator is non-zero there). At 0 the formula gives $0/0$, so we need the limit: $\lim_{x\to 0} \dfrac{\sin x}{x} = 1$ — the number from the table in Lecture 1. Continuity at 0 therefore forces $\boxed{c = 1}$. (With $c = 1$ this function is continuous on all of $\mathbb{R}$; it is the function whose graph you met again in Lecture 6, crossing its horizontal asymptote $y = 0$ for ever.)

Problem 5 *Limits by continuity. Evaluate, and name what makes each step legal.*

(a) $\lim_{x\to 2} \dfrac{x^3-4x+1}{x^2+1}$

(b) $\lim_{x\to 3} \sqrt{x^2+7}$

(c) $\lim_{x\to 1} \sin\!\left(\dfrac{\pi x}{2}\right)$

(d) $\lim_{x\to 4} \sqrt{\dfrac{x^2-16}{x-4}}$

(e) $\lim_{x\to 2} \cos\!\left(\pi\,\dfrac{x^2-4}{x-2}\right)$

Solution.

(a) The function is rational and $x=2$ is in its domain (the denominator x^2+1 is never 0), so it is continuous at 2 and "limit" means "substitute": $\lim_{x\to 2} \dfrac{x^3-4x+1}{x^2+1} = \dfrac{2^3-4\cdot 2+1}{2^2+1} = \dfrac{8-8+1}{5} = \dfrac{1}{5}$.

(b) x^2+7 is a polynomial, so $x^2+7 \to 3^2+7 = 16$; the square root is continuous at 16, so the limit passes inside it: $\lim_{x\to 3} \sqrt{x^2+7} = \sqrt{\lim_{x\to 3}(x^2+7)} = \sqrt{16} = 4$.

(c) $\dfrac{\pi x}{2} \to \dfrac{\pi}{2}$ and sin is continuous everywhere, so $\lim_{x\to 1} \sin\dfrac{\pi x}{2} = \sin\!\left(\lim_{x\to 1} \dfrac{\pi x}{2}\right) = \sin\dfrac{\pi}{2} = 1$.

(d) Two steps, in this order. First the inside, where substitution gives $0/0$: for $x \neq 4$, $\dfrac{x^2-16}{x-4} = \dfrac{(x-4)(x+4)}{x-4} = x+4$, so the inside tends to 8. Then the root, continuous at 8: $\lim_{x\to 4} \sqrt{\dfrac{x^2-16}{x-4}} = \sqrt{8} = 2\sqrt{2}$.

(e) Again inside first: for $x \neq 2$, $\pi \cdot \dfrac{x^2-4}{x-2} = \pi \cdot \dfrac{(x-2)(x+2)}{x-2} = \pi(x+2) \to 4\pi$. Then cos is continuous at 4π, so the limit passes inside: $\lim_{x\to 2} \cos\!\left(\pi\,\dfrac{x^2-4}{x-2}\right) = \cos(4\pi) = 1$.

Problem 6 *The Intermediate Value Theorem.*

In (a)–(d), show that the equation has a solution in the given interval. Write the argument in full: name the function, say why it is continuous, give the two values, and state the conclusion.

(a) $x^3+2x-4=0$ on $(1,2)$

(b) $x^4+x-3=0$ on $(1,2)$

(c) $\cos x = x$ on $(0,1)$

(d) $\sqrt{x+1} = x^2$ on $(1,2)$

(e) True or false: "$f(0)=-1$ and $f(2)=3$, so $f(c)=0$ for some c in $(0,2)$." Justify, or give a counterexample.

Solution.

(a) Let $f(x) = x^3+2x-4$; the equation is $f(x)=0$. f is a polynomial, so it is continuous on $\mathbb{R}$ and in particular on the closed interval $[1,2]$. $f(1) = 1+2-4 = -1 < 0$ and $f(2) = 8+4-4 = 8 > 0$. The number 0 lies between -1 and 8, so by the Intermediate Value Theorem there is a number c in $(1,2)$ with $f(c)=0$: the equation has a solution there. □

(b) Let $f(x) = x^4+x-3$, a polynomial and therefore continuous on $[1,2]$. $f(1) = 1+1-3 = -1 < 0$, $f(2) = 16+2-3 = 15 > 0$. Since 0 lies between -1 and 15, the IVT gives a c in $(1,2)$ with $f(c)=0$. □

(c) An equation of the form left = right becomes an equation "= 0" by subtracting: let $g(x) = \cos x - x$, so that $\cos x = x$ means $g(x)=0$. cos is continuous everywhere and x is a polynomial, so the difference g is continuous on $[0,1]$. $g(0) = \cos 0 - 0 = 1 > 0$ and $g(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.46 < 0$. By the IVT there is c in $(0,1)$ with $g(c)=0$, that is $\cos c = c$. □

(d) Let $h(x) = \sqrt{x+1} - x^2$; the equation is $h(x)=0$. On $[1,2]$ we have $x+1 \geq 2 > 0$, so $\sqrt{x+1}$ is defined and continuous there; x^2 is a polynomial; a difference of continuous functions is continuous. $h(1) = \sqrt{2}-1 \approx 0.41 > 0$ and $h(2) = \sqrt{3}-4 \approx -2.27 < 0$. By the IVT there is c in $(1,2)$ with $h(c)=0$, that is $\sqrt{c+1} = c^2$. □

(e) False as stated. The Intermediate Value Theorem has one hypothesis, and it is missing here: nothing says that f is continuous. A counterexample is the step $f(x) = \begin{cases} -1, & x < 1, \\ 3, & x \geq 1, \end{cases}$ which satisfies $f(0) = -1$ and $f(2) = 3$ and never takes the value 0 — it jumps straight over it. If "f is continuous on $[0, 2]$" is added, the statement becomes true and is exactly the IVT with $N = 0$.

Part II · Limits at infinity and horizontal asymptotes (Section 3.4) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. If $\lim_{x\to\infty} f(x) = 3$, then $y = 3$ is a horizontal asymptote of the graph of f.
2. The graph of a function can never cross its horizontal asymptote.
3. A function can have two different horizontal asymptotes.
4. The graph of a rational function can have two different horizontal asymptotes.
5. $\lim_{x\to\infty}(x^2 - x) = 0$, because $\infty - \infty = 0$.
6. $\displaystyle\lim_{x\to-\infty} \frac{1}{x^3} = 0$.
7. For $x < 0$, $\sqrt{x^2} = x$.
8. $\displaystyle\lim_{x\to\infty} \sin x = 0$.
9. If $\lim_{x\to\infty} f(x) = \infty$, then the graph of f has no horizontal asymptote as $x \to \infty$.
10. $\displaystyle\lim_{x\to\infty} \left(\sqrt{x^2 + 3x} - x\right) = 0$.

Solution.

1 True. That is precisely the definition: the line $y = L$ is called a horizontal asymptote of f when $f(x) \to L$ as $x \to \infty$, or when $f(x) \to L$ as $x \to -\infty$. Nothing more is required.

2 False. An asymptote says where the graph is heading, not where it may not go. The graph of $\dfrac{\sin x}{x}$ has the horizontal asymptote $y = 0$ and crosses it at every multiple of π — infinitely often.

3 True. $g(x) = \dfrac{x}{\sqrt{x^2+1}}$ tends to 1 as $x \to \infty$ and to -1 as $x \to -\infty$, because $\sqrt{x^2} = |x|$ changes sign with x. Two ends, two asymptotes.

4 False. For a rational function the two ends always agree: dividing by the highest power of x in the denominator gives the same number at $+\infty$ and at $-\infty$ (the ratio of the leading coefficients when the degrees are equal, 0 when the top degree is smaller, and no finite limit at all when it is larger). So a rational function has at most one horizontal asymptote. The example in item 3 is not rational — there is a square root in it.

5 False. "$\infty - \infty$" is not a number and decides nothing; it is an instruction to rewrite. Here $x^2 - x = x(x-1)$, and both factors run to $+\infty$, so $\lim_{x\to\infty}(x^2 - x) = \infty$.

6 True. As $x \to -\infty$ the number $|x|^3$ grows without bound, so its reciprocal tends to 0 (from below, since $x^3 < 0$). This is the basic limit $1/x^r \to 0$ with $r = 3$.

7 False — and this is the trap of the whole section. $\sqrt{x^2}$ is by definition the *non-negative* number whose square is x^2, that is $|x|$. For $x < 0$ that number is $-x$, not x. Check with $x = -3$: $\sqrt{9} = 3 = -(-3)$.

8 False. As x grows, $\sin x$ keeps sweeping through every value between -1 and 1 and never settles, so the limit does not exist. (Compare $\dfrac{\sin x}{x}$, whose amplitude shrinks to 0.)

9 True. A horizontal asymptote $y = L$ would mean the values stay as close to L as we like from some point on. But $f(x) \to \infty$ means the values eventually exceed every number, so they leave the band from $L - \varepsilon$ to $L + \varepsilon$ and never come back.

10 False. Both terms run to infinity, so "$\infty - \infty$": rewrite. Multiplying by the conjugate, $\sqrt{x^2+3x} - x = \dfrac{(x^2+3x) - x^2}{\sqrt{x^2+3x} + x} = \dfrac{3x}{\sqrt{x^2+3x} + x} = \dfrac{3}{\sqrt{1 + 3/x} + 1} \longrightarrow \dfrac{3}{2}$.

Problem 7 *Rational functions at infinity. Divide by the highest power of x in the denominator.*

(a) $\lim_{x\to\infty} \dfrac{4x^2-3x+1}{2x^2+5}$

(b) $\lim_{x\to\infty} \dfrac{2x+7}{3x^2-x}$

(c) $\lim_{x\to\infty} \dfrac{x^3+1}{4x^2+x}$

(d) $\lim_{x\to-\infty} \dfrac{5x^3-2x}{1-3x^3}$

(e) $\lim_{x\to-\infty} \dfrac{x^2+1}{x^3-x}$

Solution.

(a) Substitution gives "∞/∞", which is not a number — rewrite first. The highest power in the denominator is x^2, so divide both parts by x^2 (legal for $x \neq 0$, and the limit at infinity uses only such x): $\lim_{x\to\infty} \dfrac{4x^2-3x+1}{2x^2+5} = \lim_{x\to\infty} \dfrac{4-3/x+1/x^2}{2+5/x^2} = \dfrac{4-0+0}{2+0} = 2$. Every term of the form $1/x^r$ dies; what survives is the ratio of the leading coefficients.

(b) Highest power in the denominator: x^2. $\lim_{x\to\infty} \dfrac{2x+7}{3x^2-x} = \lim_{x\to\infty} \dfrac{2/x+7/x^2}{3-1/x} = \dfrac{0+0}{3-0} = 0$. The top has smaller degree than the bottom, so the quotient is squeezed to 0: $y = 0$ is a horizontal asymptote.

(c) Highest power in the denominator: x^2 again. $\dfrac{x^3+1}{4x^2+x} = \dfrac{x+1/x^2}{4+1/x}$. Now the numerator tends to ∞ while the denominator tends to 4, so the quotient grows without bound: the limit is ∞. The top degree is larger than the bottom, so there is *no* horizontal asymptote in this direction.

(d) Highest power in the denominator: x^3. $\lim_{x\to-\infty} \dfrac{5x^3-2x}{1-3x^3} = \lim_{x\to-\infty} \dfrac{5-2/x^2}{1/x^3-3} = \dfrac{5-0}{0-3} = -\dfrac{5}{3}$. The same answer comes out at $+\infty$: a rational function heads for the same level at both ends.

(e) Highest power in the denominator: x^3. $\lim_{x\to-\infty} \dfrac{x^2+1}{x^3-x} = \lim_{x\to-\infty} \dfrac{1/x+1/x^3}{1-1/x^2} = \dfrac{0+0}{1-0} = 0$.

Problem 8 *Square roots at infinity: the sign trap and the conjugate. (Write $\sqrt{x^2} = |x|$ before you divide.)*

(a) $\lim_{x\to\infty} \dfrac{\sqrt{9x^2+2}}{4x-1}$

(b) $\lim_{x\to-\infty} \dfrac{\sqrt{9x^2+2}}{4x-1}$

(c) $\lim_{x\to\infty} \left(\sqrt{x^2+5x}-x\right)$

(d) $\lim_{x\to\infty} \left(\sqrt{x^2+3x}-\sqrt{x^2-x}\right)$

(e) $\lim_{x\to-\infty} \left(x+\sqrt{x^2+4x}\right)$

Solution.

(a) Here $x \to \infty$, so $x > 0$ and $\sqrt{x^2} = |x| = x$. Pull it out of the root: $\sqrt{9x^2+2} = \sqrt{x^2}\sqrt{9+2/x^2} = x\sqrt{9+2/x^2}$. Dividing top and bottom by x, $\dfrac{\sqrt{9x^2+2}}{4x-1} = \dfrac{\sqrt{9+2/x^2}}{4-1/x} \longrightarrow \dfrac{\sqrt{9}}{4} = \dfrac{3}{4}$.

(b) Same function, other end — and now the sign matters. For $x \to -\infty$ we have $x < 0$, so $\sqrt{x^2} = |x| = -x$ and $\sqrt{9x^2+2} = -x\sqrt{9+2/x^2}$. Dividing by x, $\dfrac{\sqrt{9x^2+2}}{4x-1} = \dfrac{-\sqrt{9+2/x^2}}{4-1/x} \longrightarrow -\dfrac{3}{4}$. Two different horizontal asymptotes, $y = \frac{3}{4}$ and $y = -\frac{3}{4}$. Sanity check: for large negative x the top is positive and the bottom negative, so the quotient must be negative.

(c) Both terms run to ∞: "$\infty - \infty$", an instruction to rewrite. Multiply and divide by the conjugate $\sqrt{x^2+5x}+x$ (positive for large x, so nothing is lost): $\sqrt{x^2+5x}-x = \dfrac{(x^2+5x)-x^2}{\sqrt{x^2+5x}+x} = \dfrac{5x}{\sqrt{x^2+5x}+x}$. Now divide top and bottom by $x > 0$, remembering $\sqrt{x^2+5x} = x\sqrt{1+5/x}$: $= \dfrac{5}{\sqrt{1+5/x}+1} \longrightarrow \dfrac{5}{1+1} = \dfrac{5}{2}$.

(d) Again $\infty - \infty$; the conjugate of a difference of two roots is their sum: $\sqrt{x^2+3x} - \sqrt{x^2-x} = \dfrac{(x^2+3x)-(x^2-x)}{\sqrt{x^2+3x}+\sqrt{x^2-x}} = \dfrac{4x}{\sqrt{x^2+3x}+\sqrt{x^2-x}}$. Divide by $x > 0$, using $\sqrt{x^2+3x} = x\sqrt{1+3/x}$ and $\sqrt{x^2-x} = x\sqrt{1-1/x}$: $= \dfrac{4}{\sqrt{1+3/x}+\sqrt{1-1/x}} \longrightarrow \dfrac{4}{1+1} = 2$.

(e) At $-\infty$ the first term goes to $-\infty$ and the root to $+\infty$: again a difference in disguise. Multiply by the conjugate $x - \sqrt{x^2+4x}$: $x + \sqrt{x^2+4x} = \dfrac{x^2-(x^2+4x)}{x-\sqrt{x^2+4x}} = \dfrac{-4x}{x-\sqrt{x^2+4x}}$. Now the sign trap: $x < 0$, so $\sqrt{x^2+4x} = |x|\sqrt{1+4/x} = -x\sqrt{1+4/x}$, and the denominator is $x + x\sqrt{1+4/x} = x\big(1+\sqrt{1+4/x}\big)$. Hence $\dfrac{-4x}{x\big(1+\sqrt{1+4/x}\big)} = \dfrac{-4}{1+\sqrt{1+4/x}} \longrightarrow \dfrac{-4}{1+1} = -2$.

Problem 9 *Find all asymptotes, horizontal and vertical. (A hole is not an asymptote.)*

(a) $y = \dfrac{3x+1}{x-2}$

(b) $y = \dfrac{x^2-1}{x^2-4}$

(c) $y = \dfrac{x-3}{x^2-9}$

(d) $y = \dfrac{x}{\sqrt{x^2+1}}$

(e) $y = \dfrac{x^2+x}{x-1}$

Solution.

(a) *Vertical:* the denominator vanishes at $x = 2$ and the numerator does not ($3 \cdot 2 + 1 = 7$), so the quotient blows up: $x = 2$ is a vertical asymptote. *Horizontal:* divide by x, $\dfrac{3+1/x}{1-2/x} \to 3$ at both ends, so $y = 3$.

(b) *Vertical:* $x^2 - 4 = 0$ at $x = \pm 2$, and the numerator $x^2 - 1$ is 3 there, not 0: two vertical asymptotes, $x = 2$ and $x = -2$. *Horizontal:* $\dfrac{1-1/x^2}{1-4/x^2} \to 1$, so $y = 1$ (the graph crosses it nowhere here, but it would be allowed to).

(c) First simplify, and say where you may: for $x \neq 3$, $\dfrac{x-3}{(x-3)(x+3)} = \dfrac{1}{x+3}$. *Vertical:* only $x = -3$. At $x = 3$ the limit is $\dfrac{1}{6}$, a finite number — a hole, not an asymptote; "denominator zero" was where to look, not the verdict. *Horizontal:* $\dfrac{1}{x+3} \to 0$ at both ends, so $y = 0$.

(d) *Vertical:* none — $x^2 + 1 \geq 1$, so the denominator is never 0 and the function is defined and continuous on all of $\mathbb{R}$. *Horizontal:* for $x > 0$, $\sqrt{x^2+1} = x\sqrt{1+1/x^2}$, so the quotient is $\dfrac{1}{\sqrt{1+1/x^2}} \to 1$; for $x < 0$, $\sqrt{x^2+1} = -x\sqrt{1+1/x^2}$, so it is $\dfrac{-1}{\sqrt{1+1/x^2}} \to -1$. Two horizontal asymptotes: $y = 1$ and $y = -1$.

(e) *Vertical:* $x = 1$ (the numerator is 2 there, not 0). *Horizontal:* dividing by x gives $\dfrac{x+1}{1-1/x}$, whose numerator grows without bound; the limit is ∞ as $x \to \infty$ and $-\infty$ as $x \to -\infty$. So there is no horizontal asymptote: when the top degree exceeds the bottom degree the graph never levels off.

Problem 10 *Infinite limits at infinity: what does the function do far out?*

(a) $\lim_{x\to\infty}\left(x^3-4x^2\right)$

(b) $\lim_{x\to-\infty}\left(x^3-4x^2\right)$

(c) $\lim_{x\to\infty}\left(x^2-x^4\right)$

(d) $\lim_{x\to-\infty}\left(2x^5+x^2\right)$

(e) $\lim_{x\to\infty}\left(\sqrt{x}-x\right)$

Solution. All but (b) are an "$\infty-\infty$" and need rewriting; in (b) both terms run to $-\infty$, so nothing is indeterminate there — but one move settles all five. Factor out the highest power: it is the term that decides the outcome, and what is left in the bracket tends to its coefficient.

(a) $x^3-4x^2=x^3\left(1-\dfrac{4}{x}\right)$. As $x\to\infty$: $x^3\to\infty$ and $1-4/x\to1$, so the product tends to ∞.

(b) Same factorisation, other end: as $x\to-\infty$, $x^3\to-\infty$ while $1-4/x\to1$, so the product tends to $-\infty$. (An odd power changes sign with x; the $-4x^2$ never gets a say.)

(c) $x^2-x^4=x^4\left(\dfrac{1}{x^2}-1\right)$. Here $x^4\to\infty$ and the bracket tends to -1, so the limit is $-\infty$: for large x the fourth power wins and drags the whole expression down.

(d) $2x^5+x^2=x^5\left(2+\dfrac{1}{x^3}\right)$. As $x\to-\infty$, $x^5\to-\infty$ and the bracket tends to 2, so the limit is $-\infty$.

(e) $\sqrt{x}-x=x\left(\dfrac{1}{\sqrt{x}}-1\right)$ for $x>0$. The bracket tends to -1 and $x\to\infty$, so the limit is $-\infty$. In words: far out, x grows faster than $\sqrt{x}$, so the difference is dominated by $-x$.

Problem 11 *Oscillation at infinity: when does the Squeeze Theorem help?*

(a) $\lim_{x\to\infty}\dfrac{\cos x}{x}$

(b) $\lim_{x\to\infty}\cos x$

(c) $\lim_{x\to\infty}\dfrac{x+\sin x}{x}$

(d) $\lim_{x\to\infty}\dfrac{\sin(x^2)}{\sqrt{x}}$

Solution.

(a) Bounded over unbounded. Since $|\cos x|\le1$, for $x>0$ we have $-\dfrac{1}{x}\le\dfrac{\cos x}{x}\le\dfrac{1}{x}$. Both bounds tend to 0, so by the Squeeze Theorem $\lim_{x\to\infty}\dfrac{\cos x}{x}=0$. The oscillation never stops, but its amplitude $1/x$ dies.

(b) Does not exist. There is nothing to squeeze: the amplitude is a fixed 1. Whatever number L one proposes, the values $\cos x$ keep returning to 1 and to -1, so they cannot stay near L. (There is no horizontal asymptote either.)

(c) Split the fraction first: $\dfrac{x+\sin x}{x}=1+\dfrac{\sin x}{x}$. By the same squeeze as in (a), $\dfrac{\sin x}{x}\to0$, so the limit is $1+0=1$. (So $y=1$ is a horizontal asymptote, and the graph crosses it infinitely often — at every $x=n\pi$.)

(d) The inside of the sine is wild, but that does not matter: whatever x is, $|\sin(x^2)|\le1$. Hence for $x>0$, $-\dfrac{1}{\sqrt{x}}\le\dfrac{\sin(x^2)}{\sqrt{x}}\le\dfrac{1}{\sqrt{x}}$, and both bounds tend to 0 (the basic limit with $r=\frac{1}{2}$). By the Squeeze Theorem the limit is 0.

Problem 12 *The precise definition at infinity.* Recall: $\lim_{x\to\infty} f(x) = L$ *means: for every* $\varepsilon > 0$ *there is a number* N *such that if* $x > N$ *then* $|f(x) - L| < \varepsilon$.

Prove each of the following from the definition. Scratch work first (find N), then the proof written forward.

(a) $\lim_{x\to\infty} \frac{1}{x} = 0$

(b) $\lim_{x\to\infty} \frac{1}{\sqrt{x}} = 0$

(c) $\lim_{x\to\infty} \frac{3x+1}{x} = 3$

(d) $\lim_{x\to\infty} x^2 = \infty$ (for every $M > 0$ there is N with $x > N \Rightarrow x^2 > M$)

Solution.

(a) *Scratch.* For $x > 0$, $\left|\frac{1}{x} - 0\right| = \frac{1}{x}$, and $\frac{1}{x} < \varepsilon$ exactly when $x > \frac{1}{\varepsilon}$. So $N = \frac{1}{\varepsilon}$ should work. *Proof.* Let $\varepsilon > 0$ be given. Take $N = \frac{1}{\varepsilon}$. If $x > N$, then $x > \frac{1}{\varepsilon} > 0$, so $\left|\frac{1}{x} - 0\right| = \frac{1}{x} < \varepsilon$. Hence $\lim_{x\to\infty} \frac{1}{x} = 0$. □

(b) *Scratch.* For $x > 0$, $\left|\frac{1}{\sqrt{x}} - 0\right| = \frac{1}{\sqrt{x}}$, and $\frac{1}{\sqrt{x}} < \varepsilon \iff \sqrt{x} > \frac{1}{\varepsilon} \iff x > \frac{1}{\varepsilon^2}$. *Proof.* Let $\varepsilon > 0$. Take $N = \frac{1}{\varepsilon^2}$. If $x > N$ then $\sqrt{x} > \frac{1}{\varepsilon}$, so $\frac{1}{\sqrt{x}} < \varepsilon$, that is $\left|\frac{1}{\sqrt{x}} - 0\right| < \varepsilon$. Hence the limit is 0. □

(c) *Scratch.* The distance to the target simplifies before anything else: for $x > 0$, $\left|\frac{3x+1}{x} - 3\right| = \left|3 + \frac{1}{x} - 3\right| = \frac{1}{x}$, which is $< \varepsilon$ exactly when $x > \frac{1}{\varepsilon}$. *Proof.* Let $\varepsilon > 0$. Take $N = \frac{1}{\varepsilon}$. If $x > N$ then $\left|\frac{3x+1}{x} - 3\right| = \frac{1}{x} < \varepsilon$. Hence $\lim_{x\to\infty} \frac{3x+1}{x} = 3$. □

(d) Here the target is ∞, so the challenge is a number M to exceed rather than a band to enter. *Scratch.* For $x > 0$, $x^2 > M \iff x > \sqrt{M}$. *Proof.* Let $M > 0$ be given. Take $N = \sqrt{M}$. If $x > N$ then $x > \sqrt{M} > 0$, so $x^2 > M$. Hence $\lim_{x\to\infty} x^2 = \infty$. □ Notice the grammar is the one you already know: *for every* challenge *there is* a response, and the response is allowed to depend on the challenge.

Challenge *(for those who finish early, or at home)*

(a) Show that every polynomial of *odd* degree has at least one real zero. (Hint: what are the limits at $+\infty$ and at $-\infty$? Then apply the Intermediate Value Theorem on a large enough interval.)

(b) Let f be continuous on $[0,1]$ with $0 \le f(x) \le 1$ for every x. Show that f has a *fixed point*: some c in $[0,1]$ with $f(c) = c$. (Hint: apply the IVT to $g(x) = f(x) - x$.)

Solution.

(1) Write $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with n odd and $a_n \neq 0$. We may assume $a_n > 0$: if not, work with $-p$, which has the same zeros.

The two ends. Factor out the leading power, as in Problem 10: $p(x) = x^n\left(a_n + \frac{a_{n-1}}{x} + \cdots + \frac{a_0}{x^n}\right)$, and the bracket tends to $a_n > 0$ at both ends. Since n is odd, $x^n \to +\infty$ as $x \to +\infty$ and $x^n \to -\infty$ as $x \to -\infty$. Hence $\lim_{x\to\infty} p(x) = +\infty$ and $\lim_{x\to-\infty} p(x) = -\infty$.

Two values of opposite sign. Because $p(x) \to +\infty$, there is a number b with $p(b) > 0$; because $p(x) \to -\infty$, there is a number $a < b$ with $p(a) < 0$.

The theorem. p is a polynomial, hence continuous on the closed interval $[a,b]$, and 0 lies between $p(a) < 0$ and $p(b) > 0$. By the Intermediate Value Theorem there is c in (a,b) with $p(c) = 0$. □

(Even degree is genuinely different: $p(x) = x^2 + 1$ has no real zero. The oddness is what makes the two ends disagree in sign.)

(2) The trick is to turn "$f(c) = c$" into "$g(c) = 0$". Put $g(x) = f(x) - x$; as a difference of continuous functions, g is continuous on $[0,1]$.

The two values. $g(0) = f(0) - 0 = f(0) \geq 0$, because f takes values in $[0, 1]$. And $g(1) = f(1) - 1 \leq 0$, for the same reason.

Two cases. If $g(0) = 0$ then $f(0) = 0$ and $c = 0$ is a fixed point; if $g(1) = 0$ then $c = 1$ is one. Otherwise $g(0) > 0$ and $g(1) < 0$, so 0 lies strictly between them, and the Intermediate Value Theorem gives a c in $(0, 1)$ with $g(c) = 0$, that is $f(c) = c$. $\square$

(In a picture: the graph of f lives inside the unit square and joins the left edge to the right edge, so it must cross the diagonal $y = x$. The IVT is what turns that picture into a proof.)